$$L = 0.2 \text{ H}$$
$$X_L = 2\pi f L = 2\pi \times 50 \times 0.2 = 62.83 \ \Omega$$
$$V_m = V_{rms}\sqrt{2} = 150\sqrt{2} = 212.13 \text{ V}$$
$$I_m = \frac{V_m}{X_L} = \frac{212.13}{62.83} = 3.38 \text{ A}$$
$$v = V_m \sin 2\pi f t = 212.13 \sin 2\pi \times 50 \times t = 212.13 \sin 100\pi t$$
$$i = I_m \sin(2\pi f t - 90°) = 3.38 \sin(100\pi t - 90°)$$

Example 3

An inductive coil having negligible resistance and 0.1 henry inductance is connected across a 200 V, 50 Hz supply. Find (i) inductive reactance, (ii) rms value of current, (iii) power, (iv) power factor, and (v) equations for voltage and current.

Solution

$$L = 0.1 \text{ H}$$
$$V = 200 \text{ V}$$
$$f = 50 \text{ Hz}$$

(i) Inductive reactance

$$X_L = 2\pi f L = 2\pi \times 50 \times 0.1 = 31.42 \ \Omega$$

(ii) rms value of current

$$I = \frac{V}{X_L} = \frac{200}{31.42} = 6.37 \text{ A}$$

(iii) Power

Since the current lags behind the voltage by 90° in purely inductive circuit, $\phi = 90°$

$$P = VI\cos\phi = 200 \times 6.37 \times \cos(90°) = 0$$

(iv) Power factor

$$\text{pf} = \cos\phi = \cos(90°) = 0$$

(v) Equations for voltage and current

$$V_m = \sqrt{2}\,V = \sqrt{2} \times 200 = 282.84 \text{ V}$$
$$I_m = \sqrt{2}\,I = \sqrt{2} \times 6.37 = 9 \text{ A}$$
$$\omega = 2\pi f = 2\pi \times 50 = 314.16 \text{ rad/s}$$
$$v = V_m \sin \omega t = 282.84 \sin 314.16\,t$$
$$i = I_m \sin\left(\omega t - \frac{\pi}{2}\right) = 9 \sin\left(314.16\,t - \frac{\pi}{2}\right)$$

Example 4

The voltage and current through circuit elements are

$v = 100 \sin(314\,t + 45°)$ *volts*

$i = 10 \sin(314\,t + 315°)$ *amperes*

(i) Identify the circuit elements. (ii) Find the value of the elements. (iii) Obtain an expression for power.

Solution $v = 100 \sin (314\,t + 45°)$

$i = 10 \sin (314\,t + 315°)$

$= 10 \sin (314\,t + 315° - 360°)$

$= 10 \sin (314\,t - 45°)$

(i) Identification of elements

From voltage and current equations, it is clear that the current i lags behind the voltage by 90°. Hence, the circuit element is an inductor.

(ii) Value of elements

$$X_L = \frac{V}{I} = \frac{V_m}{I_m} = \frac{100}{10} = 10\ \Omega$$

$$X_L = \omega L$$

$$10 = 314\,L$$

$$L = 31.8 \text{ mH}$$

(iii) Expression for power

$$p = -\frac{V_m I_m}{2} \sin 2\omega t = -\frac{100 \times 10}{2} \sin (2 \times 314 t) = -500 \sin 628\,t$$

Example 5

A capacitor has a capacitance of 30 microfarads which is connected across a 230 V, 50 Hz supply. Find (i) capacitive reactance, (ii) rms value of current, (iii) power, (iv) power factor, and (v) equations for voltage and current.

Solution $C = 30\ \mu F$

$V = 230$ V

$f = 50$ Hz

(i) Capacitive reactance

$$X_C = \frac{1}{2\pi f C} = \frac{1}{2\pi \times 50 \times 30 \times 10^{-6}} = 106.1\ \Omega$$

(ii) rms value of current

$$I = \frac{V}{X_C} = \frac{230}{106.1} = 2.17 \text{ A}$$

(iii) Power

Since the current leads the voltage by 90° in purely capacitive circuit, $\phi = 90°$

$$P = VI \cos \phi = 230 \times 2.17 \times \cos (90°) = 0$$

(iv) Power factor

$$\text{pf} = \cos \phi = \cos (90°) = 0$$

(v) Equations for voltage and current

$$V_m = \sqrt{2}\, V = \sqrt{2} \times 230 = 325.27 \text{ V}$$

$$I_m = \sqrt{2}\, I = \sqrt{2} \times 2.17 = 3.07 \text{ A}$$

$$\omega = 2\pi f = 2\pi \times 50 = 314.16 \text{ rad/s}$$
$$v = V_m \sin \omega t = 325.27 \sin 314.16\, t$$
$$i = I_m \sin\left(\omega t + \frac{\pi}{2}\right) = 3.07 \sin\left(314.16\, t + \frac{\pi}{2}\right)$$

Example 6

Find the rms value of current flowing through a 314 μF capacitor when connected to a 230 V, 50 Hz, 1φ AC supply. **[Dec 2015]**

Solution

$$C = 314\ \mu\text{F}$$
$$V = 230 \text{ V}$$
$$f = 50 \text{ Hz}$$
$$X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi \times 50 \times 314 \times 10^{-6}} = 10.14\ \Omega$$
$$I = \frac{V}{X_C} = \frac{230}{10.14} = 22.68 \text{ A}$$

4.4 SERIES R-L CIRCUIT

Figure 4.13 shows a pure resistor R connected in series with a pure inductor L across an alternating voltage v.

Let V and I be the rms values of applied voltage and current.

Potential difference across the resistor = $V_R = RI$

Potential difference across the inductor = $V_L = X_L I$

The voltage $\bar{V}_R$ is in phase with the current $\bar{I}$ whereas the voltage $\bar{V}_L$ leads the current $\bar{I}$ by 90°.

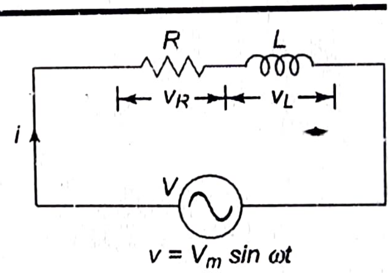

Fig. 4.13 Series R-L circuit

Phasor Diagram

Steps for drawing phasor diagram

1. Since the same current flows through series circuit, $\bar{I}$ is taken as reference phasor.
2. Draw $\bar{V}_R$ in phase with $\bar{I}$.
3. Draw $\bar{V}_L$ such that it leads $\bar{I}$ by 90°.
4. Add $\bar{V}_R$ and $\bar{V}_L$ by triangle law of vector addition such that
$$\bar{V} = \bar{V}_R + \bar{V}_L$$
5. Mark the angle between $\bar{I}$ and $\bar{V}$ as ϕ.

The phasor diagram is shown in Fig. 4.14.